

IMPORTANT FACTS

We are supposed to find the day of the week on a given date.

For this, we use the concept of *odd days*.

I. Odd Days: In a given period, the number of days more than the complete weeks are called *odd days*.

II. Leap Year:

(i) Every year divisible by 4 is a leap year, if it is not a century.

(ii) Every 4th century is a leap year and no other century is a leap year.

Note : A leap year has 366 days.

Examples:

(i) Each of the years 1948, 2004, 1676 etc. is a leap year.

(ii) Each of the years 400, 800, 1200, 1600, 2000 etc. is a leap year.

(iii) None of the years 2001, 2002, 2003, 2005, 1800, 2100 is a leap year.

III. Ordinary Year:

The year which is not a leap year is called an ordinary year. An ordinary year has 365 days.

IV. Counting of Odd Days:

(i) 1 ordinary year = 365 days = (52 weeks + 1 day).

$\therefore$ 1 ordinary year has 1 odd day.

(ii) 1 leap year = 366 days = (52 weeks + 2 days).

$\therefore$ 1 leap year has 2 odd days.

(iii) 100 years = 76 ordinary years + 24 leap years
 $= (76 \times 1 + 24 \times 2)$ odd days = 124 odd days
 $= (17 \text{ weeks} + 5 \text{ days}) \equiv 5$ odd days.

$\therefore$ Number of odd days in 100 years = 5

Number of odd days in 200 years = $(5 \times 2) \equiv 3$ odd days

Number of odd days in 300 years = $(5 \times 3) \equiv 1$ odd day.

Number of odd days in 400 years = $(5 \times 4 + 1) \equiv 0$ odd day.

Similarly, each one of 800 years, 1200 years, 1600 years, 2000 years, etc. has 0 odd days.

V. Day of the Week Related to Odd Days:

No. of days	0	1	2	3	4	5	6
Day	Sun.	Mon.	Tues.	Wed.	Thurs.	Fri.	Sat.

SOLVED EXAMPLES

Ex. 1. What was the day of the week on 16th July, 1776?

Sol. 16th July, 1776 = (1775 years + Period from 1.1.1776 to 16.7.1776)

Counting of odd days:

Number of odd days in 1600 years = 0

Number of odd days in 100 years = 5

75 years = 18 leap years + 57 ordinary years

$= (18 \times 2 + 57 \times 1)$ odd days = 93 odd days

$= (13 \text{ weeks} + 2 \text{ days}) \equiv 2$ odd days

$\therefore$ 1775 years have

$= (0 + 5 + 2)$ odd days $= 7$ odd days $\equiv 0$ odd day.

Jan. Feb. March April May June July
 $(31 + 29 + 31 + 30 + 31 + 30 + 16) = 198$ days

198 days $= (28 \text{ weeks} + 2 \text{ days}) \equiv 2$ odd days.

$\therefore$ Total number of odd days $= (0 + 2) = 2$.

Hence, the required day is Tuesday.

Ex. 2. *What was the day of the week on 15th August, 1947?*

Sol. 15th August, 1947 $= (1946 \text{ years} + \text{Period from 1.1.1947 to 15.8.1947})$

Odd days in 1600 years $= 0$

Odd days in 300 years $= (5 \times 3) = 15 \equiv 1$

46 years $= (11 \text{ leap years} + 35 \text{ ordinary years})$

$= (11 \times 2 + 35 \times 1)$ odd days $= 57$ odd days

$= (8 \text{ weeks} + 1 \text{ day}) \equiv 1$ odd day.

$\therefore$ Odd days in 1946 years $= (0 + 1 + 1) = 2$.

Jan. Feb. March April May June July Aug
 $(31 + 28 + 31 + 30 + 31 + 30 + 31 + 15) = 227$ days

227 days $= (32 \text{ weeks} + 3 \text{ days}) \equiv 3$ odd days.

Total number of odd days $= (2 + 3) = 5$.

Hence, the required day is Friday.

Ex. 3. *What was the day of the week on 4th June, 2002?*

Sol. 4th June, 2002 $= (2001 \text{ years} + \text{Period from 1.1.2002 to 4.6.2002})$

Odd days in 1600 years $= 0$

Odd days in 400 years $= 0$

Odd days in 1 ordinary year $= 1$

Odd days in 2001 years $= (0 + 0 + 1) = 1$

Jan. Feb. March April May June
 $(31 + 28 + 31 + 30 + 31 + 4) = 155$ days

$= 22 \text{ weeks} + 1 \text{ day} \equiv 1$ odd day

Total number of odd days $= (1 + 1) = 2$

$\therefore$ Required day is Tuesday.

Ex. 4. *On what dates of March 2005 did Friday fall?*

Sol. First we find the day on 1.3.2005

1.3.2005 $= (2004 \text{ years} + \text{Period from 1.1.2005 to 1.3.2005})$

Odd days in 1600 years $= 0$ Odd days in 400 years $= 0$

4 years $= (1 \text{ leap year} + 3 \text{ ordinary years})$

$= (1 \times 2 + 3 \times 1)$ odd days $= 5$ odd days

Jan. Feb. March
 $(31 + 28 + 11)$

$= 60$ days $= (8 \text{ weeks} + 4 \text{ days}) \equiv 4$ odd days.

Total number of odd days $= (0 + 0 + 5 + 4) = 9 \equiv 2$ odd days

$\therefore$ 1.3.2005 was Tuesday. So, Friday lies on 4.3.2005

Hence, Friday lies on 4th, 11th, 18th and 25th of March, 2005.

Ex. 5. *Prove that the calendar for the year 2003 will serve for the year 2014.*

Sol. We must have same day on 1.1.2003 and 1.1.2014.

So, number of odd days between 31.12.2002 and 31.12.2013 must be 0.

This period has 3 leap years and 8 ordinary years.

Number of odd days $= (3 \times 2 + 8 \times 1) = 14 \equiv 0$ odd day

$\therefore$ Calendar for the year 2003 will serve for the year 2014.

EXERCISE

(OBJECTIVE TYPE QUESTIONS)

Directions: Mark (✓) against the correct answer in each of the following:

- January 1, 2007 was Monday. What day of the week lies on Jan. 1, 2008?
(a) Monday (b) Tuesday
(c) Wednesday (d) Sunday
- January 1, 2008 is Tuesday. What day of the week lies on Jan. 1, 2009?
(a) Monday (b) Wednesday
(c) Thursday (d) Sunday
- On 8th Dec, 2007 Saturday falls. What day of the week was it on 8th Dec. 2006?
(a) Sunday (b) Thursday
(c) Tuesday (d) Friday
- On 6th March, 2005 Monday falls. What was the day of the week on 6th March, 2004?
(a) Sunday (b) Saturday
(c) Tuesday (d) Wednesday
- The calendar for the year 2007 will be the same for the year:
(a) 2014 (b) 2016
(c) 2017 (d) 2018
- On what dates of April, 2001 did Wednesday fall?
(a) 1st, 8th, 15th, 22nd, 29th
(b) 2nd, 9th, 16th, 23rd, 30th
(c) 3rd, 10th, 17th, 24th
(d) 4th, 11th, 18th, 25th
- What was the day of the week on 17th June, 1998?
(a) Monday (b) Tuesday
(c) Wednesday (d) Thursday
- What was the day of the week on 28th May, 2006?
(a) Thursday (b) Friday
(c) Saturday (d) Sunday
- What will be the day of the week on 15th August, 2010?
(a) Sunday (b) Monday
(c) Tuesday (d) Friday
- Today is Monday. After 61 days, it will be
(a) Wednesday (b) Saturday
(c) Tuesday (d) Thursday
- The last day of a century cannot be
(a) Monday (b) Wednesday
(c) Tuesday (d) Friday
- Which of the following is not a leap year?
(a) 700 (b) 800
(c) 1200 (d) 2000
- How many days are there in x weeks x days?
(a) $7x^2$ (b) $8x$
(c) $14x$ (d) 7
- It was Sunday on Jan 1, 2006. What was the day of the week on Jan 1, 2010?
(a) Sunday (b) Saturday
(c) Friday (d) Wednesday
- On 8th Feb, 2005 it was Tuesday. What was the day of the week on 8th Feb, 2004?
(a) Tuesday (b) Monday
(c) Sunday (d) Wednesday
- For a certain month, the dates of three of the Sundays are even numbers. Then, the 15th of the that month falls on a [SSC—CGL (Tier I) Exam, 2012]
(a) Thursday (b) Friday
(c) Saturday (d) Sunday
- What was the day of the week on 15 August, 1947?
[DMRC—Customer Relationship Assistant (CRA) Exam, 2016]
(a) Saturday (b) Friday
(c) Thursday (d) Wednesday
- The calendar for the year 2009 will be the same as that of the year [DMRC—Customer Relationship Assistant (CRA) Exam, 2016]
(a) 2013 (b) 2014
(c) 2015 (d) 2014

ANSWERS

1. (b) 2. (c) 3. (d) 4. (b) 5. (d) 6. (d) 7. (c) 8. (d) 9. (a) 10. (b)
11. (c) 12. (a) 13. (b) 14. (c) 15. (c) 16. (c) 17. (b) 18. (c)

SOLUTIONS

1. The year 2007 is an ordinary year. So, it has 1 odd day.
1st day of the year 2007 was Monday.
1st day of the year 2008 will be 1 day beyond Monday.
Hence, it will be Tuesday.
2. The year 2008 is a leap year. So, it has 2 odd days.
1st day of the year 2008 is Tuesday (Given)
So, 1st day of the year 2009 is 2 days beyond Tuesday.
Hence, it will be Thursday.
3. The year 2006 is an ordinary year. So, it has 1 odd day.
So, the day on 8th Dec, 2007 will be 1 day beyond the day on 8th Dec, 2006.
But, 8th Dec, 2007 is Saturday.
 $\therefore$ 8th Dec, 2006 is Friday.
4. The year 2004 is a leap year. So, it has 2 odd days.
 $\therefore$ The day on 6th March, 2005 will be 2 days beyond the day on 6th March, 2004.
But, 6th March, 2005 is Monday.
 $\therefore$ 6th March, 2004 is Saturday.
5. Count the number of odd days from the year 2007 onwards to get the sum equal to 0 odd day.

Year	2007	2008	2009	2010	2011	2012	2013	2014	2015	2016	2017
Odd day	1	2	1	1	1	2	1	1	1	2	1

- Sum = 14 odd days $\equiv$ 0 odd day.
 $\therefore$ Calendar for the year 2018 will be the same as for the year 2007.
6. We shall find the day on 1st April, 2001.
 1st April, 2001 = (2000 years + Period from 1.1.2001 to 1.4.2001)
 Odd days in 1600 years = 0
 Odd days in 400 years = 0
 Jan. Feb. March April
 (31 + 28 + 31 + 1) = 91 days $\equiv$ 0 odd days.
 16. The dates of three of the Sundays are even number is 2, 9, 16, 23, 30.
 So, on 16th of that month = Sunday, 15th of that month falls on a Saturday

17. 15 August, 1947 means 1946 complete years + first 7 months upto July 1947 + 15 days of August 1947
 1600 years have 0 odd day.
 300 years have 1 odd day
 46 years have 11 leap years and 35 ordinary years
 $= (11 \times 2) + (35 \times 1)$
 $= 22 + 35 = 57$ odd days
 $= 8 \times 7 + 1$ odd days
 $= 8$ weeks + 1 odd day
 Up to 1946 there are 1 + 1 = 2 odd days
 January 1947 $\Rightarrow$ 3 odd days
 February 1947 $\Rightarrow$ 0 odd days
 (1947 is a normal year)
 March 1947 $\Rightarrow$ 3 odd days
 April 1947 $\Rightarrow$ 2 odd days
 May 1947 $\Rightarrow$ 3 odd days
 June 1947 $\Rightarrow$ 2 odd days
 July 1947 $\Rightarrow$ 3 odd days
 Up to 15 August $\Rightarrow$ 15 odd days
 Total number of odd days up to 15 August 1947
 $= 2 + 3 + 0 + 3 + 2 + 3 + 2 + 3 + 15 = 33$ odd days.
 Hence, 15th August, 1947 was Friday.
18. 2008 was a leap year.
 A leap year has two odd days.
 Suppose year 2008 starts with a Monday.
 Then the first day of 2009 was Wednesday.
 Now,
 First day of 2010 $\Rightarrow$ Thursday
 First day of 2011 $\Rightarrow$ Friday
 First day of 2012 $\Rightarrow$ Saturday
 Because 2012 was a leap year
 First day of 2013 $\Rightarrow$ Monday
 First day of 2014 $\Rightarrow$ Tuesday
 First day of 2015 $\Rightarrow$ Wednesday
 Thus, the calendar for the year 2009 was the same as that of the year 2015.